

Honeywell

Remote Displays

User's GuideOperations

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R400

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About This Document

This document describes how to install and configure GUS Remote Displays server and client. It also describes how to view the status of GUS Remote Displays.

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Document Category
Operations

References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Doc ID
<i>Native Window User's Guide</i>	GU20
<i>Configuration Utility User's Guide</i>	TP20X
<i>TPS System Implementation Guide</i>	TP08X
<i>Integrated Experion-TPS Nodes (ES-T, ESVT, & ACE-T) User's Guide</i>	EX23

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the Honeywell book set.
	REFERENCE - INTERNAL: Identifies an additional source of information within the Honeywell book set.

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Contents

1. Introduction

1.1 Overview

Terms and acronyms

Term	Definition
Experion Station - TPS	Experion Station – TPS (ES-T) is an operator node in the Experion system that has an LCNP board that is executing Universal Station (US) personality. For the purpose of this document, the node is capable of executing both GUS Server and GUS Client functions.
ES-C	Experion Console Station
ES-CE	Experion Station – Console Extension. For the purpose of this document, the ES-CE node executes the GUS Client, and the ES-T performs the function of the GUS Server.
GUS Alarm Summary Display	A display functionally equivalent to the US R520 Alarm Summary Display. This function is not supported in the Experion TPS (replaced by the Experion Alarm Summary display).
GUS Client (GC)	A computer connected to a single GUS Server via a communication link and capable of hosting the Client components that is required to achieve Remote Displays.
GUS Faceplate	An OCX representing one “slot” of the GUS Group Display.
GUS Group Display	A display functionality equivalent to the US R520 Group Display.
GUS Remote Cluster	The set of computers consisting of a single GUS Server and its associated GUS Clients. In an Experion system, this consists of a single ES-T node and its associated ES-CE nodes.

1 Introduction

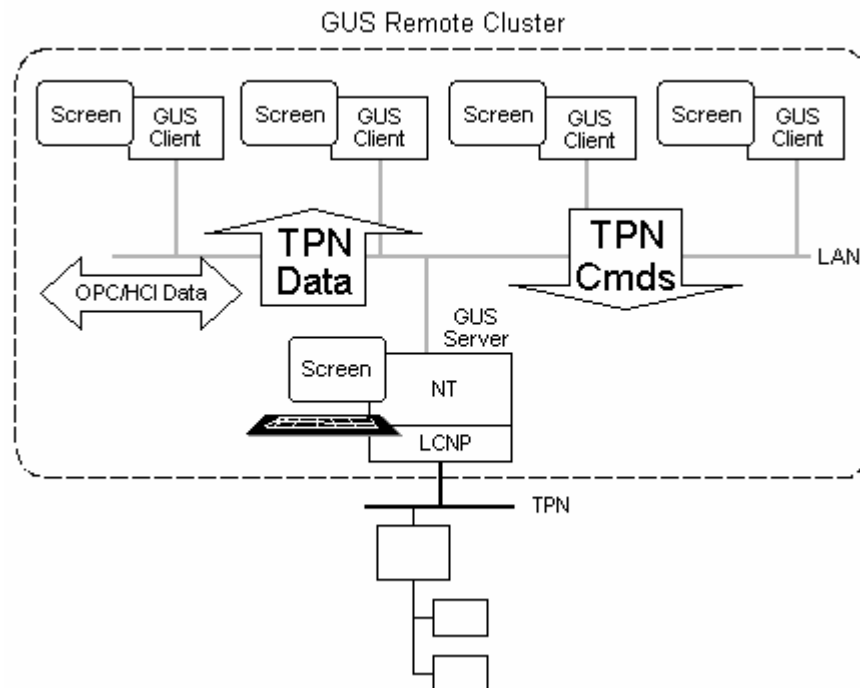
1.1 Overview

Term	Definition
GUS Server (GS)	A computer connected to the TPN through an LCNP. The LCNP executes the Universal Personality. In TPS, a GUS Server can be connected to up to four GUS Clients. In Experion, a GUS Server can be connected to up to three GUS Remote Clients (the server node includes a client as part of the GUS display package).
IKB/OEP	Acronyms for the Integrated Keyboard (IKB) and Operator Entry Panel (OEP).
TPN Access Levels	TPN Security accommodates the following levels of access to the TPN Data Owners. These are called TPN Access levels. The levels are: • Engineer • Supervisor • Operator

General description

The Global User Station (GUS) Remote Displays package provides remote access to TPN data from up to four GUS Clients in a TPS system, or up to three remote GUS clients in an Experion system, through a single GUS Server (host). The set of computers consisting of a single GUS server and its associated GUS clients is called a GUS Remote Cluster. Displays are run in the GUS clients. An Experion system must include the TPS options to support the GUS Remote Display functionality.

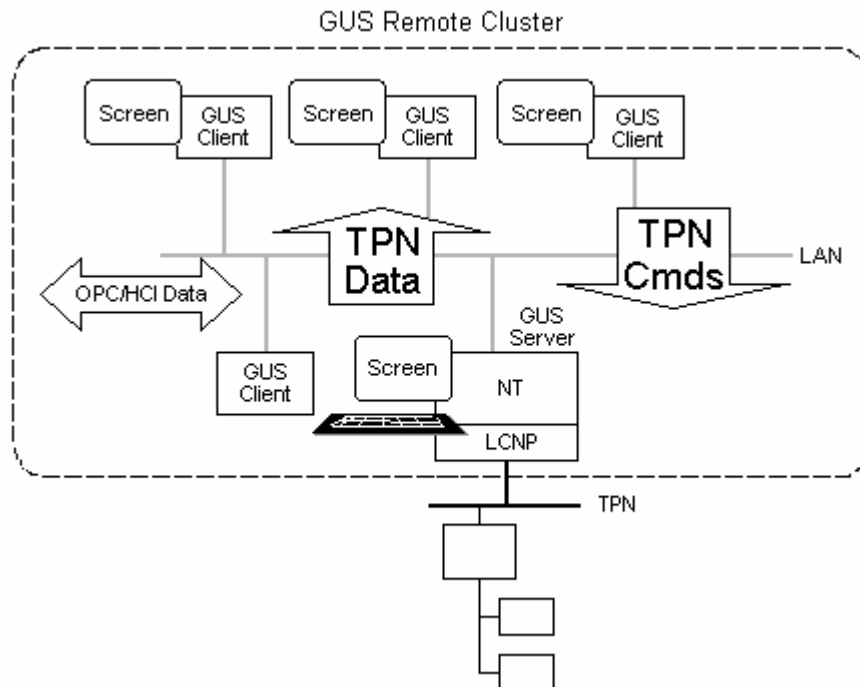
The following figure shows a GUS Remote Cluster configured with four separate clients. This is a valid configuration for a TPS system. In an Experion system, three remote GUS clients are supported.



1 Introduction

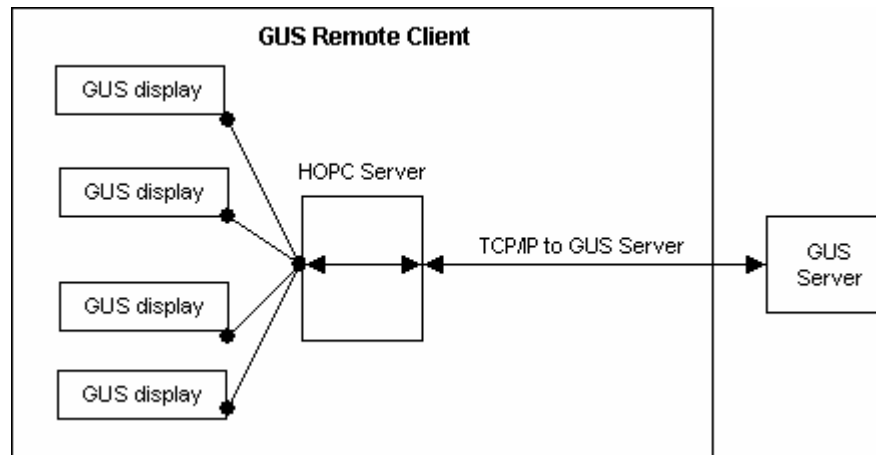
1.1 Overview

The following figure shows another way you can configure your system, by locating one of the clients on the server. This configuration is valid for a TPS system. In an Experion system, the GUS server node supports a maximum of three remote GUS clients, and the server node includes a GUS client as part of the server functionality.



GUS displays

GUS display functionality for GUS Remote Clients makes it possible to run more than one GUS display at the same time. The following figure illustrates how four GUS displays on a GUS Remote Client, through the HOPC Server, can connect to a GUS server.



ATTENTION

The Multiple Displays package is required to run more than one GUS display on a GUS Remote Client. Without this software, you will only be able to run one GUS display on each Remote Client.

Invoking GUS displays

There are several ways to run a GUS display:

1. You can invoke a GUS display using the RUNPIC.EXE application.
2. You can invoke a GUS display using script.
3. If you have installed the GUS Display Builder software, you can use Display Builder commands to invoke a GUS display.
4. You can program function keys (F1...) to invoke a GUS display using the SCHEM actor.
5. On an Experion node, you can invoke a GUS display from the Experion Station command prompt or from HMIWeb displays.

1 Introduction

1.1 Overview

Area Data Base

In a TPS system, the GUS Server supports a common Area Data Base for a GUS Remote Cluster, in which all computers in the GUS Remote Cluster do the following:

- Use the same alarm list
- Use the same button configuration
- Use the same group definitions
- Follow the same GUS Display Search path (However, each client can supplement the Area search path with a Local extension through the GUS Display Runtime Settings page in the Configuration utility.)

In an Experion system, behavior is as follows:

- The ES-T and all ES-CE computers in the cluster follow the same GUS Display Search path.
- The ES-T and all ES-CE computers in the cluster see the same TPN group definitions.
- Native Window uses the same button configuration (N/A for GUS displays on ES-CE nodes).
- The ES-T and all ES-CE computers in the cluster see the same alarm list (through Experion Alarm Summary).

Security

Computers in a GUS Remote Cluster operate independently with respect to the TPN security level (keylock). Changing the security at one computer does not change it at other computers in the same GUS Remote Cluster.

You can use the Remote Native Window to set the keylock and you must configure Signon Manager (also known as COP for Console Operator Process) to select the keylock at operator sign-on. Refer to the *TPS System Implementation Guide* for detailed instructions to configure Signon Manager keylock settings.



TIP

Keylock is not supported on Experion systems. You must use Signon Manager to change operator and associated TPN access level.

To control access to GUS Servers, during the installation and configuration procedure you should physically secure the network connecting GUS Servers to a GUS Client by using the switches and TCP/IP address filtering capabilities available in the network infrastructure. When you configure a GUS Server, you can specify which GUS Clients will be able to connect to the GUS Server by selecting the "Enable Clients" from the List radio button. Refer to The Enable Clients from List Option section in this document.

1 Introduction

1.2 GUS Remote Cluster Functionality

1.2 GUS Remote Cluster Functionality

GUS Display and Window functionalities

The following table describes the range of GUS Display and Window functionalities for a GUS Client and a GUS Server within a GUS Remote Cluster.

Function	Description
SafeView	SafeView executes on any computer in the GRC.
GUS Displays	In a TPS system, the GUS Server supports up to four GUS Clients (one of which can be located on the GUS Server). Each GUS Client is capable of executing GUS runtime and builder displays. The Multiple GUS Displays option is available. In an Experion system, the ES-T supports up to three ES-CE nodes. The ES-T and each ES-CE node are capable of executing GUS runtime and builder displays. The Multiple GUS Displays option is available.
GUS Group Display	Each GUS Client can execute only one GUS Group Display.
GUS Message Summary Display	Not supported.
GUS Alarm Summary Display	In a TPS system, the GUS Client can execute one GUS Alarm Summary Display. The GUS Alarm Summary Display is not available in an Experion system.
Faceplate Application	Each GUS Client can support only one Faceplate Application.
GUS Alarm Window OCX	In a TPS system, each GUS Client can support up to two GUS Alarm Window OCXs. The GUS Server does not impose limits on the number of GUS Alarm Window OCXs. Not available in an Experion system,
GUS Faceplate OCX	Each GUS Client can support up to four GUS Faceplate OCXs.
GUS Trend OCX	Each GUS Client can support up to four GUS Trend OCXs.
GUS Message Window OCX	Not supported.

Function	Description
Native Window Display	The Native Window can be placed on any computer in the GRC. GUS Server supports a single Native Window connection. In a TPS system, interactions such as schematic invocations between GUS runtime displays and Native Window are NOT supported. In an Experion system, interactions between GUS runtime displays and Native Window are supported on an ES-T node, even if this node is hosting remote ES-CE nodes.
(TPN) File Transfer	Available only on the GUS Server.
Integrated Keyboard (IKB) and Operator Entry Keyboard (OEP)	In a TPS system, available only on the GUS Server. No restriction on the LCN-connected Experion nodes. Generally, on the ES-T nodes, IKB/OEP can be controlled by both Experion and TPS software. On the ES-CE nodes, the IKB/OEP is under Experion control.
Emulated Bernoulli Drives	Available only on the GUS Server.
TPN Security	TPN Security level is determined locally for each GUS Client/ES-T/ES-CE.
HCI Servers	In a TPS system, not certified to run on any computer in the GRC.

1 Introduction

1.2 GUS Remote Cluster Functionality

2. System Requirements

2.1 Hardware Requirements

Computers (TPS system)

GUS Server

CPU:	GUS Product CPU
Memory:	256 Megabytes minimum for Windows 2000/XP Operating Systems
Disk:	4 Gigabytes
Screen:	Customer choice
Keyboards:	Standard engineering keyboard (IKB/OEP supported)
Additional Hardware:	LCNP4 card

Remote GUS Client

CPU:	400 MHz minimum for remote operations
Memory:	256 Megabytes minimum for Windows .2000 and Windows XP Operating Systems
Disk:	4 Gigabytes
Screen:	GUS Product Screen
Keyboards:	Standard engineering keyboard (IKB/OEP NOT supported)
Additional Hardware:	None

Computers (Experion TPS system)

Hardware requirements are defined by the Experion nodes.

2 System Requirements

2.2 Software Requirements (TPS systems)

Networking

A large number of commercially available network media can be utilized to connect GUS Clients to GUS servers. For remote operations, minimum 10 Base T network is required.

GUS Remote Displays do not require that the computers in GUS Remote Cluster belong to a common TPS domain. It is recommended that the TPS domain be configured, as it provides a platform for common operator security policies and data files replication mechanisms, among others.

For an Experion system, all LCN-connected Experion nodes should belong to the network domain. Experion High Security packages must be installed on all nodes. Users at the ES-CE nodes should belong to the appropriate local user groups that have been defined by the Experion High Security package at the host ES-T node.

GUS Remote Displays do not require that the computers in GUS Remote Cluster belong to a common Windows domain. However, in order to configure user permissions to validate the initial selection and keylock changes, the Windows users of GUS Client must be visible at the GUS Server PC. That requires that minimally one-way trust must exist between Windows domains that contain GUS Server and GUS Client PCs (the GUS Server's PC domain is trusting the domain of the GUS Client PC).

GUS Client selects GUS Server by host name or directly by IP address. If a host name is used, the DNS (Domain Name System) or Hosts file must be configured to resolve the server host name.

2.2 Software Requirements (TPS systems)

GUS Server

- Microsoft Windows 2000 or Windows XP (with Service Packs recommended from Honeywell)
- GUS Base
- GUS Remote Displays Server



ATTENTION

The Windows Recovery Disk is provided with the GUS Remote Displays Server package.

GUS Client

- Microsoft Windows 2000 or Windows XP (with Service Packs recommended from Honeywell)
- GUS Remote Displays Client
- GUS Display Builder (optional)
- GUS HCI Client-Add In (optional)
- GUS Multiple Displays (required only if you want to run more than one GUS Display at the same time)
- SafeView (optional)
- Native Window Client (optional)



ATTENTION

Other third party software can be installed if the following are true:

- It has been qualified by Honeywell to operate with the above GUS display software.
- It accesses TPN data only through the GUS display scripting engine or HCI.



REFERENCE - INTERNAL

Refer to the *TPS System Software Package Usage by Node Type* section in the *TPS System Implementation Guide* for a complete list of supported packages.

2.3 Software Requirements (Experion systems)

GUS Server (ES-T)

- Experion Station - Console
- GUS Displays Runtime
- GUS Display Builder (optional)
- GUS Multiple Displays (optional)

2 System Requirements

2.3 Software Requirements (Experion systems)

GUS Client (ES-CE)

- Experion Station - Flex
- GUS Remote Displays Client
- Experion Remote Native Window (optional)
- GUS Display Builder (optional)
- GUS Multiple Displays (optional)



ATTENTION

Other third party software can be installed if the following are true:

- It has been qualified by Honeywell to operate with the above GUS display software.
 - It accesses TPN data only through the GUS display scripting engine or HCI.
-

3. Installation and Configuration

3.1 Required Documentation



REFERENCE - INTERNAL

- [TPS System Implementation Guide](#)
 - [System Configuration Utility User's Guide](#)
 - [Integrated Experion-TPS Nodes \(ES-T, ESVT, & ACE-T\) User's Guide](#)
-

3.2 Installing a GUS Remote Displays Server (TPS System)

Prerequisites

The GUS Base Software as described in the *TPS System Implementation Guide* must be installed before you install the GUS Remote Displays Server. Depending on your system configuration (TPS domain present or not), follow the implementation procedure for a TPS GUS Node or for a Network GUS Node.



ATTENTION

The following GUS TPN backplane modules must be present and configured for this node:

- CSCHEM
 - MSCHEM
 - GUSALA (not required for Experion)
 - GUSALAG (not required for Experion)
 - UPBASE
 - YGO CX
-

3 Installation and Configuration

3.2 Installing a GUS Remote Displays Server (TPS System)

Procedure

Follow these steps to install the GUS Remote Displays Server.

Step	Action
1	Uninstall the following three packages before installing the new software: • GUS Display Builder • GUS Display HCI Client Add-in • GUS Display Runtime
2	Install the GUS Base Software and GUS Remote Displays Server software.
3	Configure a GUS Remote Displays Server, as described in the section Configuring a GUS Remote Displays Server .
	ATTENTION Installation defaults GUS Remote Server to allow connections from all clients.
4	Configure remote access permissions.
	REFERENCE INTERNAL For information, refer to the section <i>Set Security Folder Permissions</i> in the <i>Remote Access Procedures</i> section of the <i>TPS System Implementation Guide</i> .

3 Installation and Configuration

3.2 Installing a GUS Remote Displays Server (TPS System)

Step	Action
5	Configure Signon Manager (LCN Keyswitch Permissions).
	REFERENCE INTERNAL For information, refer to the <i>Configuring Signon Manager</i> section in the <i>TPS System Implementation Guide</i> .
	ATTENTION • Failure to configure the remote access and the keyswitch settings of Signon Manager will result in the VIEW ONLY operation of clients connecting to this GUS Remote Server. • In order to configure LCN keyswitch permissions, Windows must be able to recognize the users of the remote GUS Clients. If GUS Client and GUS Server PCs belong to different Windows domains, the GUS Server's domain must trust GUS Client's domain.
	REFERENCE INTERNAL If remote access by the Native Window is required, refer to the section <i>Enable Remote Native Window Sessions</i> in the Remote Access Procedures section of the <i>TPS System Implementation Guide</i> for assistance.

3 Installation and Configuration

3.2 Installing a GUS Remote Displays Server (TPS System)

Step	Action
6	Verify that the GUS Remote LXS is enabled (checked), as indicated on the TPS Configuration Utility Devices/Services page example that follows.
7	Reboot and log on as local administrator.

Devices/Services : GUS05

Devices and Services

Description
This page is used to configure the automatic startup of Device Drivers and Services.
NOTE - Changes to these settings do not take effect until the system is restarted.

TPS Admin and Operator Signon
☐ TPSadmin ☒ TPS Signon Manager Service

LCNP Device Driver and Emulation Services
☒ Device Driver (tdcaint.sys) ☒ Services (Tdcaserver)

Time Synchronization Service
☐ Enable Synchronization Interval (Minutes) 2

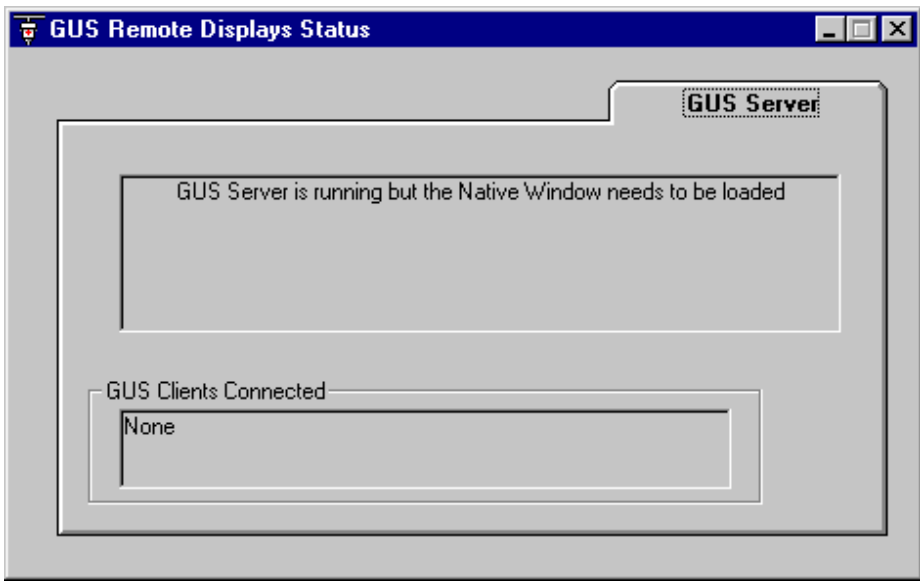
Additional Software Services
☒ LCN File Service ☒ File Transfer
☐ Directory Data Trimmer ☒ Carbon Copy
☒ GUS Remote LXS

Optional Hardware Devices
☐ Carroll Touch Screen Driver ☐ Integrated Keyboard Driver

OK Cancel Apply Help

3 Installation and Configuration

3.2 Installing a GUS Remote Displays Server (TPS System)

Step	Action
8	Check the status of the GUS Server from GUS Remote Displays Status window, GUS Server page. You can double-click the Remote Status icon  to call up the GUS Remote Displays Status window. 
9	Load the Native Window.
10	Check the GUS Remote Displays Status window, GUS Server page to verify that the GUS Server is running.
11	Optionally, install and configure the GUS Remote Displays Client (follow the desired GUS Remote Client installation procedure for your particular node type, as described in the following two sections).

3 Installation and Configuration

3.3 Installing a GUS Remote Displays Server (Experion System)

3.3 Installing a GUS Remote Displays Server (Experion System)

Procedure

Perform these steps to install and configure the GUS Remote Displays Client on an ES-T's GUS Remote Displays Server.

Step	Action
1	Insert the Integrated Experion-TPS Nodes CD in the ES-T's CD drive and install the GUS Displays Runtime package using the licensed package installer.
2	Enable remote clients using the GUS Remote Displays server configuration page.
3	Restart the computer.

3.4 Installing a GUS Remote Displays Client on a GUS Remote Displays Server (TPS System)

Prerequisites

Before installing GUS Remote Displays Client, the following software must be installed:

- GUS Remote Displays Server
- GUS Base

Procedure

Perform these steps to install and configure the GUS Remote Displays Client on a GUS Remote Displays Server.

Step	Action
1	Remove any previously installed GUS software packages.
2	Install the GUS Remote Displays Client software.

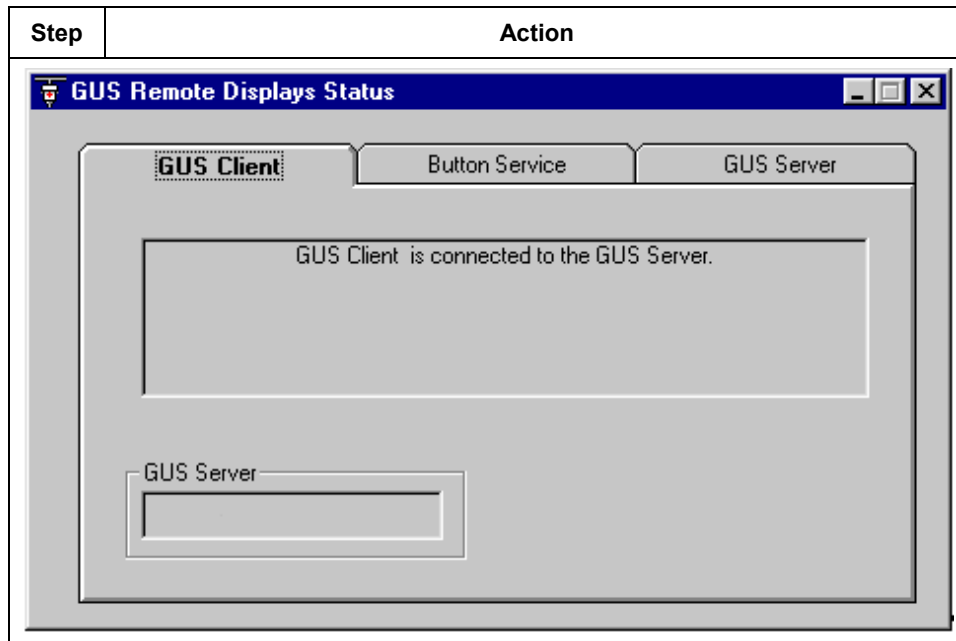
3 Installation and Configuration

3.4 Installing a GUS Remote Displays Client on a GUS Remote Displays Server (TPS System)

Step	Action
3	Optionally, install and configure other supported GUS packages. Note: "GUS Display Runtime" and "GUS Remote Displays Client" must not be installed on the same node. Refer to the <i>Installing TPS Software Packages</i> section in the <i>TPS System Implementation Guide</i>.
	ATTENTION To run more than one GUS Display at the same time, ensure that you install the GUS Multiple Displays package.
4	Configure the GUS Remote Displays Client, as described in the Configuring a GUS Remote Displays Client section. Check, and if required, change settings on the two configuration pages: • GUS Remote Displays Client (select Local Host) and • GUS Alarm/Message/Group Displays. Return to this procedure after configuring the GUS Remote Displays Client.
5	Log off, and then log on to the machine to start communications with the GUS Remote Displays Server.
6	Verify that the GUS Client and Buttonsvc are connected to the GUS Server using the GUS Remote Displays Status window. (Double-click the Remote Status icon to call it up.)

3 Installation and Configuration

3.4 Installing a GUS Remote Displays Client on a GUS Remote Displays Server (TPS System)



Special considerations

- The installation program creates three entries under All Users/ Startup menu: Button Service, GUS Picture Manager and GUS Remote Displays Status. These items activate GUS Remote Displays components. If the All Users setting is not appropriate, move these items under particular user startup menu, at the same time moving contents of the Programs/Honeywell TPS items.
- GUS displays can be located on global share or replicated into local drive. The Displays Search Path is provided by the Area database and is propagated to the GUS Remote Displays Client through the lcnfilesvc/button service connection.
- At a later time, uninstalling the Remote GUS Remote Displays Client on the GUS Remote Server Node may require reinstalling the GUS Remote Displays Server. Rather than uninstalling the GUS Remote Displays Client, disable it through the GUS Remote Displays Client page of the TPS Configuration Utility.

3.5 Installing a GUS Remote Displays Client on a Non-TPN Node (TPS System)

Prerequisites

- TCP/IP connection to GUS Remote Displays Server PC (verify by Ping command)
- If HCI Client Add-In is to be installed, the PC must be in a TPS domain. Follow the procedures for configuring the TPS Client.

Procedure

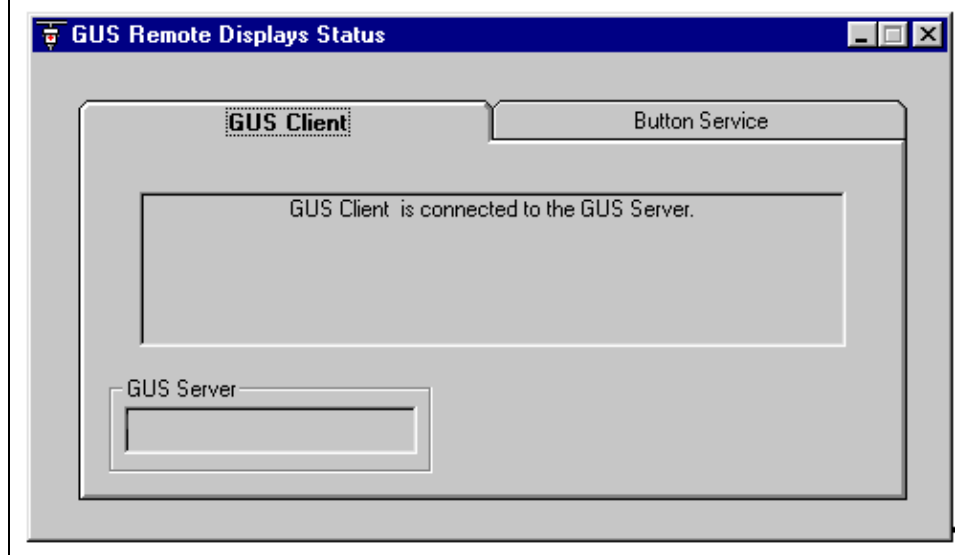
Perform these steps to install the GUS Remote Displays Client on a non-TPN node.

Step	Action
1	Determine if a GUS Remote Displays Client has already been installed. If it has not, skip steps 2 through 4 and go directly to step 5.
2	Call up the Devices/ Services page of the TPS Configuration Utility and uncheck all devices and services.
3	Perform a cold reboot of the PC and login as local administrator, holding down the Shift Key (to prevent activation of startup programs).
4	Uninstall existing GUS packages.
5	Install Microsoft Redistributable Libraries (on the TPS System Software CD).
6	Restart and log on as local administrator.
7	Install GUS Remote Displays Client software.
8	Optionally, install other supported GUS packages. Note: "GUS Display Runtime" and "GUS Remote Displays Client" must not be installed on the same node.
	REFERENCE INTERNAL Refer to the <i>TPS System Software Package Usage by Node Type</i> section in the <i>TPS System Implementation Guide</i> for a complete list of supported packages.
9	Configure the GUS Remote Displays Client, as described in the Configuring a GUS Remote Displays Client section. Check, and if required, change settings on the two configuration pages: GUS Remote Displays Client and GUS Alarm/Message/Group Displays.

3 Installation and Configuration

3.5 Installing a GUS Remote Displays Client on a Non-TPN Node (TPS System)

Step	Action
10	Log off and then log on to start communications with the GUS Remote Displays Server.
11	Check the GUS Remote Displays Status to verify that the GUS Client and Buttonsvc are connected to the GUS Server.



Special considerations

- The installation program creates three entries under All Users/ Startup menu: Button Service, GUS Picture Manager and GUS Remote Displays Status. These items activate GUS Remote Displays components. If the All Users setting is not appropriate, move these items under particular user startup menu, at the same time moving contents of the Programs/Honeywell TPS items.
- GUS displays can be located on global share or replicated into local drive. The Displays Search Path is provided by the Area database and is propagated to the GUS Remote Displays Client through the lcnfilesvc/button service connection.

3.6 Installing a GUS Remote Displays Client (Experion System)

Prerequisites

GUS software installs on top of the Experion Flex Station. Additionally, the following Common Components packages are required:

- Workstation high Security
- Signon Manager
- System Management Display (or Runtime)
- IKB Service (if IKB/OEP is present)



ATTENTION

The remote user Windows account must be a member of one of the local user groups that were created by the High Security workstation package on the ES-T computer. For example, an operator who logs on the remote computer must belong to the Local Operators group on the ES-T computer. This association comes naturally if ES-T and ES-CE computers belong to the same domain, and operator accounts are members of the global groups that were created by the Experion High Security domain package.

3 Installation and Configuration

3.7 Troubleshooting

Procedure

Step	Action
1	Install the Base Experion Flex Station /CE software. Refer to the <i>Integrated Experion-TPS Nodes (ES-T, ESVT, & ACE-T) User's Guide (EX23)</i> for Steps 1 and 2.
2	Install the required Common Components.
3	Verify that the ES-T is running with remote GUS clients enabled.
4	Insert the Integrated Experion-TPS Nodes CD in the ES-CE CD drive and install GUS Remote Displays Client package using the licensed package installer.
5	Enter and select the name of the ES-T node as the current GUS Server on the GUS Remote Displays Client configuration page and enable connection to the GUS Server. See "Configuring a GUS Remote Displays Client."
6	Log off and log on (Windows login).

3.7 Troubleshooting

Troubleshooting GUS Remote Display problems

Perform these steps to troubleshoot problems with the GUS Remote Displays Server or GUS Remote Displays Client.

Step	Action
1	Follow the recommendations on the GUS Remote Displays Status display. If there is a problem with the GUS Remote Status Display, check to make sure that the Microsoft Redistributable Libraries (present on the TPS System Software CD) have been installed.
2	Check the Windows Event Viewer log messages, considering the sources: • GusLxsClient (messages from the GUS Remote Displays Client) • GusRemoteLxs (GUS Remote Displays Server messages) • Buttonsvc (GUS Remote Displays Client – Button Service messages) • Lcnfilesvc (GUS Remote Displays Server messages) Associated sources: LcnDataServer (hopcsrvr), tdcaserver (interface to lcnp).

Step	Action
3	Verify your network setup. In particular, check the following: ping <i>server_name</i> or ping <i>server_ipaddress</i> (The IP address of the GUS Server may be used when configuring Remote GUS Client.) netstat -a (Slowness may indicate network or domain name server failure.)

3.8 Network Load

Recommended actions if network is saturated

The operation of Remote GUS Displays Server and Remote GUS Displays Clients introduces a load on the computer network. The magnitude of the load depends on the number and characteristics of invoked GUS displays (such as the number of parameters and frequency of scanning).

It is recommended that you measure this additional load, and if it appears to be significant, or if the existing traffic is close to saturating the network, use switches to isolate GUS Remote Displays PCs.

3.9 Configuring a GUS Remote Displays Client

Getting Started

The configuration (connection) of a GUS Remote Displays Client is done using the GUS Remote Displays Client page of the standard Honeywell TPS Configuration Utility.

Page overview

The GUS Remote Displays Client Page:

- Allows the GUS Client to connect to a specific GUS Server, and
- Configures GUS Servers by computer host name or IP Address.

The configuration page for GUS Remote Displays Client can be viewed after the GUS Remote Displays Client package has been installed on the node.

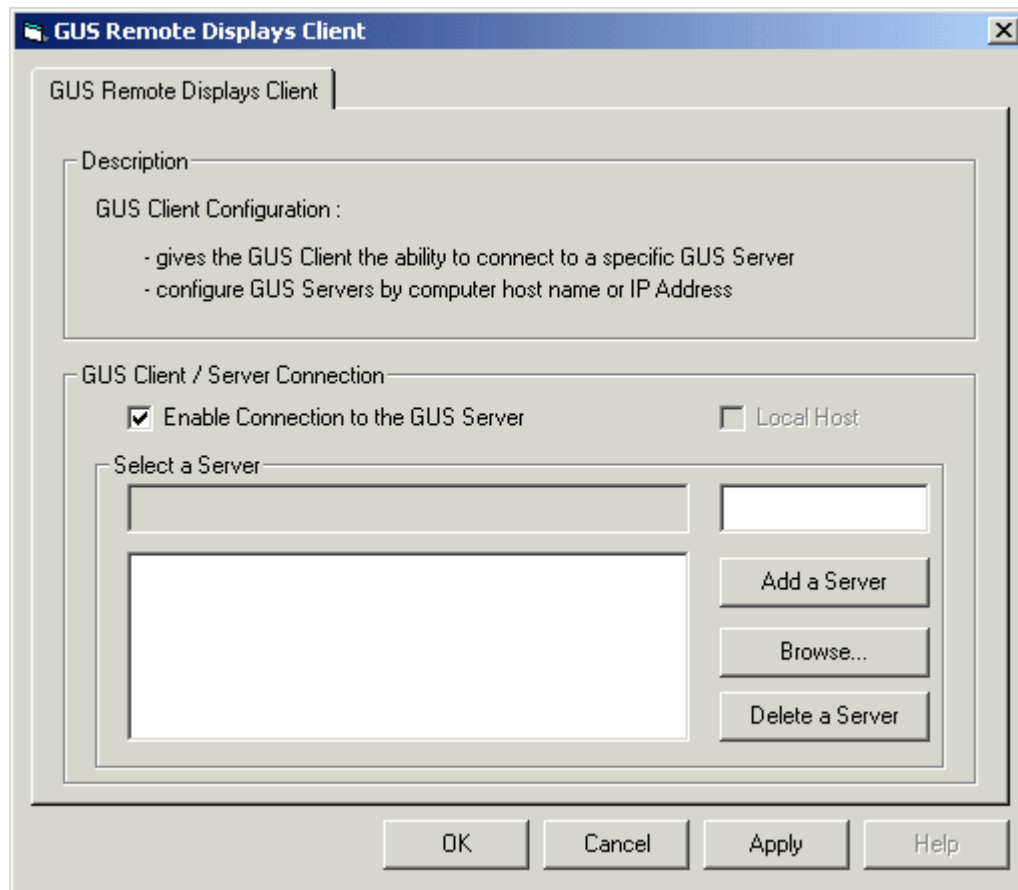
3 Installation and Configuration

3.9 Configuring a GUS Remote Displays Client

Accessing the GUS Remote Displays Client Page

Step	Action
1	Invoke the TPS Configuration Utility .
	REFERENCE – INTERNAL For instructions to invoke the TPS Configuration Utility, refer to the section <i>Starting the TPS Configuration Utility</i> in the <i>TPS System Configuration Utility User's Guide</i> . On Experion TPS nodes, the configuration utility is under Start > All Programs > Experion PKS > System Management > Configuration Utility .
2	Select the following: Configure > GUS Remote Displays Client RESULT: The following GUS Remote Displays Client page will be displayed.

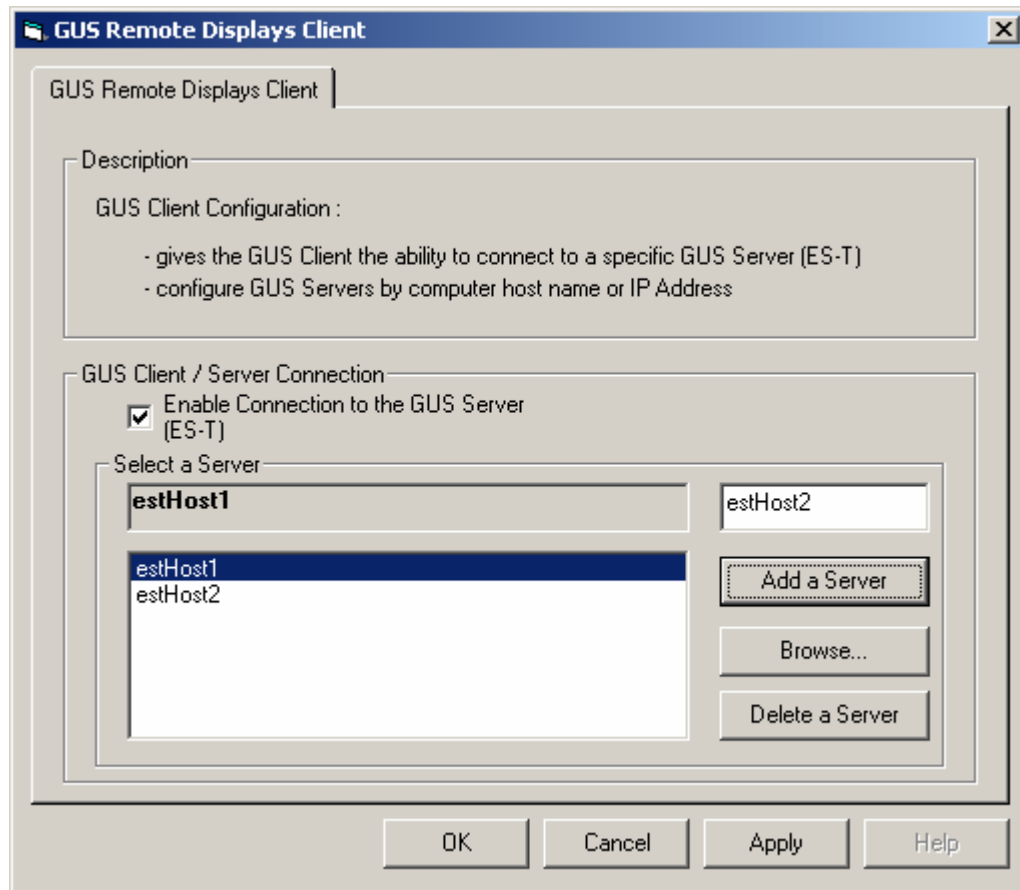
The GUS Remote Displays Client Page for TPS Systems is shown in the following figure.



The GUS Remote Displays Client Page for Experion Systems is shown in the following figure. Note that it is very similar to the TPS version, except that the Local Host check box is omitted on the Experion version. This is because on the ES-T's Remote Displays Server, the local GUS Client is always enabled and connected to the local host.

3 Installation and Configuration

3.9 Configuring a GUS Remote Displays Client



Permission to change configuration

Permission to change configuration of a GUS Remote Displays Client is granted by the ACL on the following three Windows Registry keys (write access):

\HKEY_LOCAL_MACHINE\Software\Honeywell\Remote Displays

\HKEY_LOCAL_MACHINE\Software\Honeywell\Remote Displays
\Configuration

\HKEY_LOCAL_MACHINE\Software\Honeywell\Remote Displays
\Configuration\Host List

By default, members of the Administrators group can configure a GUS Remote Displays Client. Members of the Administrators group can assign these permissions to other users by use of the REGEDT32 tool, if it is required.

GUS Client / Server connection options

Check the **Enable Connection to the GUS Server** box to connect GUS Client components to a GUS Server. The default setting is unchecked.

If the setting is unchecked, GUS Display Builder will automatically start in the “off-line” mode. This is useful when the GUS Remote Displays Client and GUS Display Builder software has been installed on a laptop computer, to allow editing of GUS displays while disconnected from the GUS Server.

Check the **Local Host** box if the GUS Client is connecting to a GUS Server installed on the same PC. The default setting is unchecked. This option is only available on TPS systems. In Experion systems, the local GUS Client is always enabled and connected to the local host.

Selecting a Remote GUS Server

Step	Action
1	Add GUS Server(s) to a list.
2	Select (highlight) GUS Server on the list.
3	Commit the change by clicking either the OK or Apply button.
	ATTENTION Committed changes will take affect <u>after</u> the next login.

3 Installation and Configuration

3.9 Configuring a GUS Remote Displays Client

Adding a GUS Server to the List

Use the following procedure or the **Browse** procedure to add an available server for the client.

Step	Action
1	Enter the server's computer hostname or IP Address in the text box.
2	Click the Add a Server button. The computer hostname or IP Address will appear in the server list.
3	Commit the change by clicking either the OK or Apply button.

The Browse Button

The Browse button procedure can be used only if the GUS Server PC is in the same Windows domain as the GUS Client. The Browse button lists all computers that belong to this domain. It does not filter out non-GUS Remote Displays PCs.

Step	Action
1	Click the Browse button.
2	Select a computer.
3	Click the OK button. The computer hostname or IP Address will appear in the server list.
4	Commit the change by clicking either the OK or Apply button.

Deleting a Server from the List

Use the following procedure to delete an available server for the client.

Step	Action
1	Select an entry in the available server list.
2	Click the Delete a Server button. The computer will be deleted from the available server list.
3	Commit the change by clicking either the OK or Apply button.

Step	Action
	ATTENTION Following deletion of the server, (re)select GUS Server for this GUS Client.

3.10 Configuring a GUS Remote Displays Server

How to get started

The configuration (connection) of a GUS Remote Displays Server is done using the GUS Remote Displays Server page of the standard Honeywell TPS Configuration Utility.

The GUS Remote Displays Server Page controls remote GUS Client connections to the GUS Server. The configuration page for a GUS Remote Displays Server can be viewed after the GUS Remote Displays Server package (TPS) or the GUS Displays Runtime package (Experion) has been installed on the server node.

Accessing the GUS Remote Displays Server page

Step	Action
1	Invoke the TPS Configuration Utility.
	REFERENCE – INTERNAL For instructions to invoke the TPS Configuration Utility, refer to the section <i>Starting the TPS Configuration Utility</i> in the <i>TPS System Configuration Utility User's Guide</i> . On Experion TPS nodes, the configuration utility is under Start > All Programs > Experion PKS > System Management > Configuration Utility .
2	Select the following from the Configuration Utility: Configure > GUS Remote Displays Server RESULT: The GUS Remote Displays Server page will be displayed.

Permission to change configuration

The permission to change configuration of GUS Remote Displays Server is granted by ACL on the following two Windows Registry keys (write access):

\HKEY_LOCAL_MACHINE\Software\Honeywell\Remote Displays
\Configuration

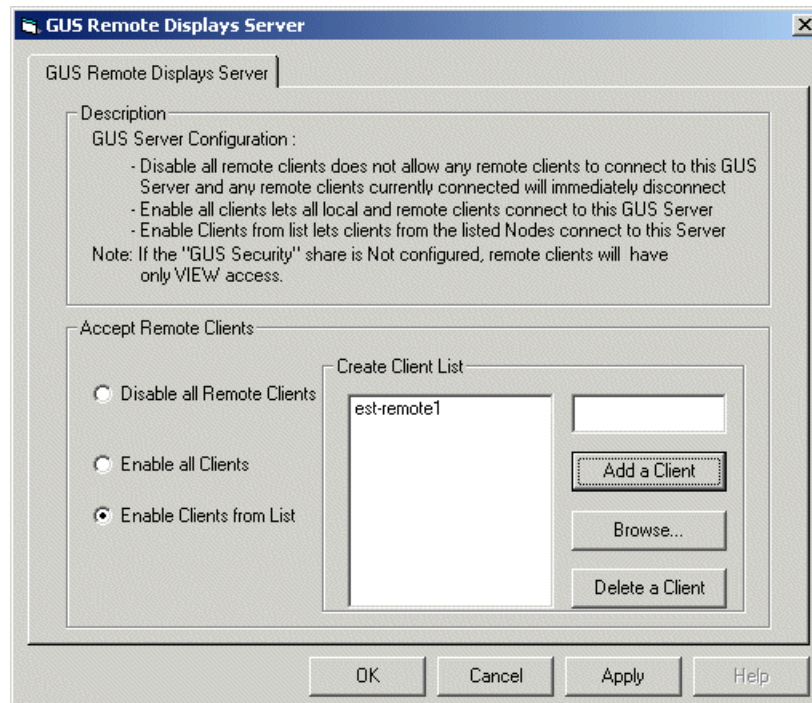
3 Installation and Configuration

3.10 Configuring a GUS Remote Displays Server

\HKEY_LOCAL_MACHINE\Software\Honeywell\Remote Displays
\Configuration\Client List

By default, members of the Administrators group can configure the GUS Remote Displays Client. Members of Administrators group can assign these permissions to other users by use of the REGEDT32 tool if it is required.

The GUS Remote Displays Server Configuration Page is shown in the following figure. This display is the same for both TPS and Experion systems.



TIP

Installation on an Experion node creates the "Gus Security" share and assigns the default access permissions to the members of the local user groups that have been created by the Workstation High Security package.

Accept Remote Clients option group



ATTENTION

On Experion TPS nodes, you must reboot the node if you make any changes to any options in this group.

The Disable All Clients Option

Select the **Disable all Clients** radio button to disallow all remote GUS Clients from connecting to the GUS Server. The default setting is unchecked. In an Experion system, the ES-T node executes the standard GUS runtime if this option is selected. Commit the change by clicking the **Apply** button.



ATTENTION

Any remote clients that were connected before the Disable All Clients option is applied will be disconnected immediately.

The Enable All Clients Option

Select the **Enable all Clients** radio button to allow all GUS Clients to connect to the GUS Server. The default setting is checked. In an Experion system, if any remote client is enabled, the ES-T will activate remote GUS displays components. If remote GUS displays are enabled, then GUS Multiple Displays option on the ES-T node will follow the rules of GUS remote displays clients, limiting the number of GUS display processes at this node to a maximum of five.

Changing the Accept Remote Clients selection must be followed by a reboot of the ES-T node, which is prompted after the selection of **OK** or **Apply**.

The Enable Clients from List Option

Select the **Enable Clients from List** radio button to specify which specific GUS Clients can connect to the GUS Server. The default setting is unchecked. Use the following procedure to specify GUS Clients.

3 Installation and Configuration

3.10 Configuring a GUS Remote Displays Server



ATTENTION

- The local computer host name must be included on the list in order to accept connection from the local GUS Client.
 - Any clients that have been removed from the list and were connected before the changes were applied will be disconnected immediately.
-

Adding a Client to the List

Use the following procedure or the **Browse** procedure to add a client.

Step	Action
1	Enter the client's computer name in the text box. (See attention below)
	ATTENTION The GUS Client Computer IP address cannot be used.
2	Click the Add a Client button. The computer name will appear in the client list.
3	Commit the change by clicking either the OK or Apply button.

The Browse Button

The Browse button procedure can be used only if the GUS Client PC is in the same Windows domain as the GUS Server. The Browse button lists all computers that belong to this domain. It does not filter out non-GUS Remote Displays PCs.

Step	Action
1	Click the Browse button.
2	Select a computer.
3	Click the OK button. The computer name will appear in the client list.
4	Commit the change by clicking either the OK or the Apply button.

The Delete a Client Button

Use this procedure to delete a client.

Step	Action
1	Select an entry in the client list.
2	Click the Delete a Client button. The computer will be deleted from the available server list.
3	Commit the change by clicking either the OK or the Apply button.

3.11 GPB Process Pool Size

Overview

The GPB (GUS Picture Builder) Process Pool is a dynamic list of the GPB processes that are currently running. It is used to provide efficient display call up by eliminating the need to start a new GPB process when a new display is invoked.

The Process Pool Setting, in conjunction with the SafeView screen configuration, determines the number of concurrent GUS displays that can be supported on a GUS Remote Displays Client. The number of concurrent GUS displays also depends on whether or not the Multiple Displays package is installed.

The following two tables show the relationship between the number of process pool entries and the maximum number of concurrent displays for systems with and without the Multiple Displays package installed.

Systems with multiple displays

The following table shows the relationship between the number of process pool entries and the maximum number of concurrent displays for a system with the Multiple Displays package installed.

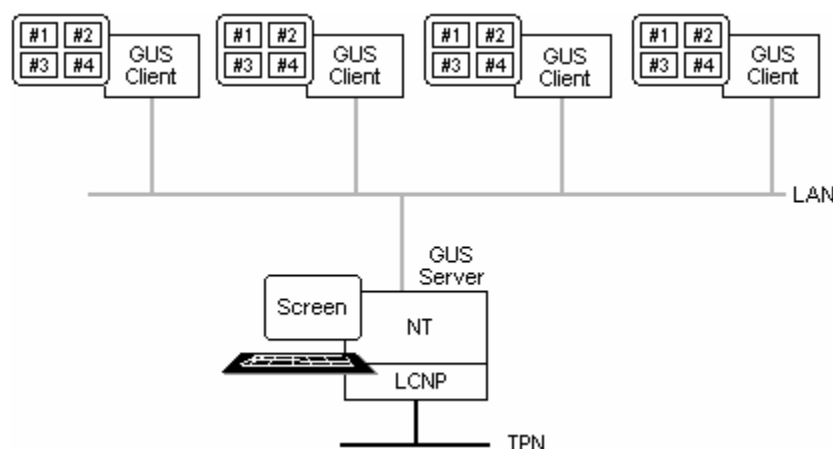
Platform	Minimum/Maximum Number of GPB Processes in the Process Pool	Default Number of GPB Processes in the Process Pool	Maximum Number of Concurrent Displays
GUS Remote Displays Client	2/5	5	4

3 Installation and Configuration

3.11 GPB Process Pool Size

The following figure shows four GUS remote clients with

- each client set up with five processes in the Process Pool, and
- each screen constrained by SafeView to manage four displays.



ATTENTION

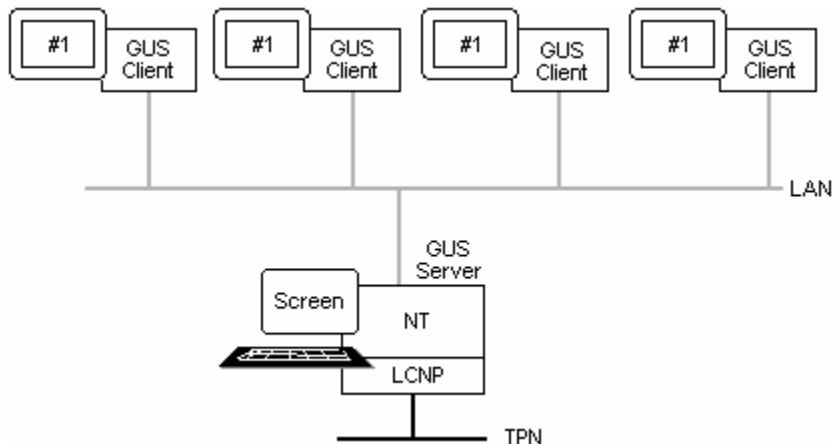
Multiple display replacement on GUS remote clients may be delayed because GPB processes are forced to disconnect from and connect to the HOPC Server.

Systems without multiple displays

The following table shows the relationship between the number of process pool entries and the recommended number of displays for a system without Multiple Displays.

Platform	Minimum/Maximum Number of GPB Processes in the Process Pool	Default Number of GPB Processes in the Process Pool	Number of Concurrent Displays
GUS Remote Displays Client	Not configurable	2	1

The following figure shows 4 GUS remote clients with each client set up with 2 processes in the Process Pool.



Configuring GUS displays for runtime



REFERENCE - INTERNAL

For details on the configuration procedure, refer to section *Configuring GUS Displays for Runtime* in the *TPS System Configuration Utility User's Guide*.

3 Installation and Configuration

3.11 GPB Process Pool Size

4. GUS Remote Displays Status

4.1 Overview

Viewing current status

The current operating status of GUS Remote Displays on a GUS is displayed by the

- GUS Remote Displays Status Icon

The status of GUS Remote Displays is displayed in the GUS Remote Displays Status window, which is comprised of three different pages. Using this window, you can

- View GUS Client Status
- View GUS Server Status
- View GUS Button Server Status (N/A on Experion nodes)

GUS Remote Displays events are captured in the Windows Event Log

4.2 GUS Remote Displays Status Icon

How to interpret status icon

The current operating status of GUS Remote Displays on a GUS is displayed by the color of the GUS Remote Displays Status icon. This icon is located in the status tray on the right side of the Windows status bar that normally resides at the bottom of your PC screen. Three different statuses that can be displayed:

- OK
- Non-fatal Error
- Fatal Error

OK Status

When the GUS Remote Displays Status is OK, the icon is a steady green color.



4 GUS Remote Displays Status

4.3 View GUS Client Status

Non-fatal error status

When the GUS Remote Displays Status has non-fatal errors, the color of the icon is a steady green color, with a yellow background. Usually, this status indicates trouble in button service.



Fatal error status

When the GUS Remote Displays Status has fatal errors, the color of the icon blinks, between a red and gray color.



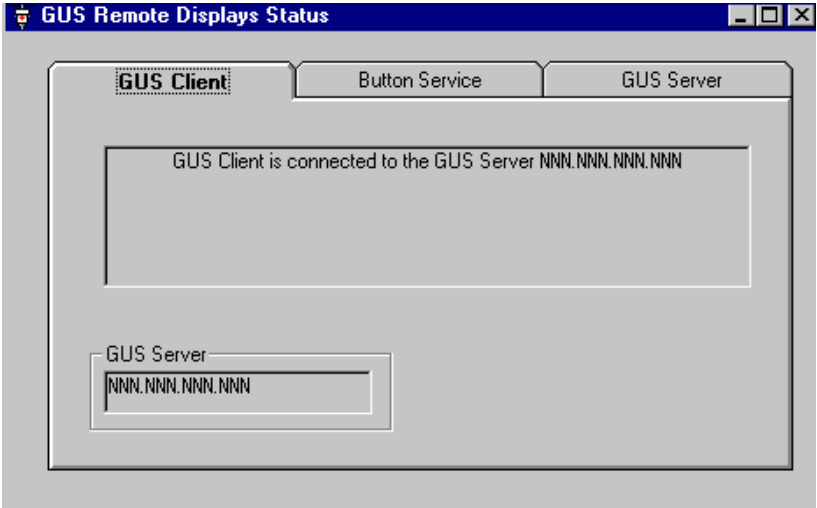
4.3 View GUS Client Status

Procedure to view GUS Client status using GUS Remote Displays icon

Step	Action
1	Left-click the GUS Remote Displays Status icon  . The GUS Remote Displays Status window appears.
2	Click the GUS Client tab. The GUS Client page is displayed.

4 GUS Remote Displays Status

4.3 View GUS Client Status

Step	Action
	
	ATTENTION The Button Service tab is not present in Experion systems.

Status messages

The two types of GUS Client status messages are configuration and run-time or connection messages. Each message explains the status of the GUS Client and tells you what to do to try and solve any existing problem.

Example - GUS Client configuration message:

GUS Client is disabled and cannot connect to the remote server. Run the TPS configuration program to enable the GUS Client.

Example - GUS Client run-time message:

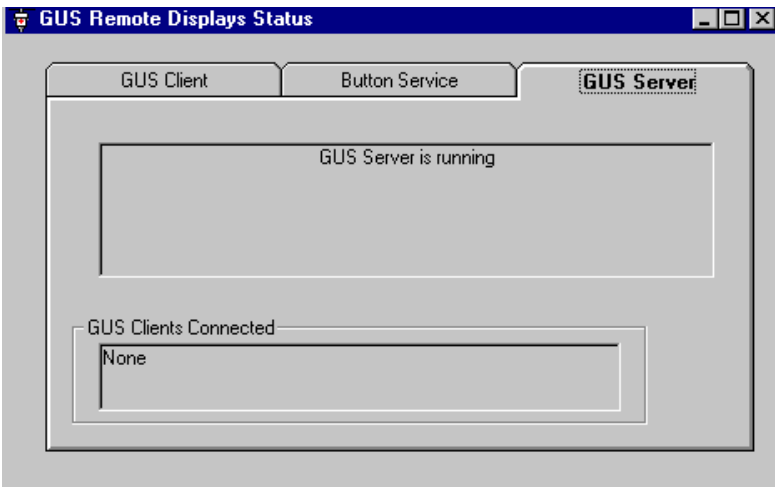
GUS Client is connected to the GUS Server but the Native Window needs to be loaded.

4.4 View GUS Server Status

Prerequisites

The GUS Server page shown below can only be displayed on a GUS that has the GUS Remote Displays Server installed.

Procedure to view GUS Server status using GUS Remote Displays icon

Step	Action
1	Left-click the GUS Remote Displays Status icon  . The GUS Remote Displays Status window appears.
2	Click the GUS Server tab. The GUS Server page is displayed.
	
	ATTENTION The Button Service tab is not present in Experion systems.

Status Messages

A number of different status messages can be displayed. Each message explains the status of the GUS Server and lists the clients that are currently connected to that server.

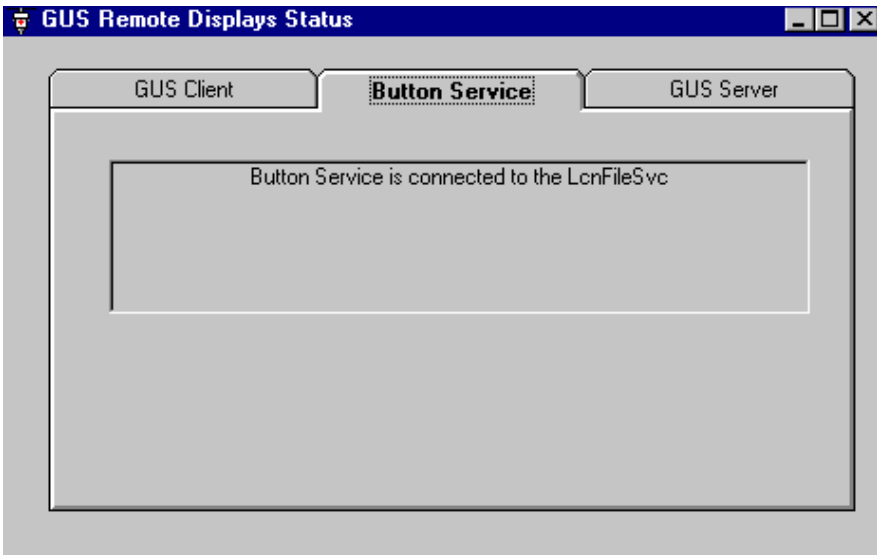
An example of a GUS Server status message is as follows: GUS Server is running.

4.5 View GUS Button Service Status (TPS Systems Only)

Button Service is a GUS Client component that works with the LCN File Service server component to propagate an Area Database change to all the GUS Clients in a GUS Remote Cluster.

Procedure to view GUS Button Service status using GUS Remote Displays icon

Step	Action
1	Left-click the GUS Remote Displays Status icon  . The GUS Remote Displays Status window appears.
2	Click the Button Service tab. The Button Service page is displayed.



4 GUS Remote Displays Status

4.6 Windows Event Log

Status messages

A number of different Button Service status messages can be displayed. The status messages let you know if the Button service is currently connected to the LCNFileSvc at the GUS Server, or if there is a problem.

An example of a Button Service status message is as follows:

- Button Service has lost connection to the LCN File Service. The user must log off and log back on to restore the connection. See the Windows Event Log for more details.

4.6 Windows Event Log

Viewing GUS Remote Displays Client initialization and connection failures

GUS Remote Display initialization and connection events are recorded in the Windows Event Log. Some typical entries include the following:

- GUS Remote Displays. Client initialization has failed because of configuration error on the client. Error value is
- GUS Remote Displays. Client connected to GUS Server *serverhostname*.
- GUS Remote Displays. Client lost connection to the GUS Server *serverhostname*.

5. GUS Display Functions

5.1 GUS Display Functions Overview

How status of GUS Display Functions is displayed

There are three ways that Display functions can work when they are executed from a GUS within a GUS Remote Cluster:

- Functions that work normally
- Functions that work but require additional actions.
- Functions that are not available

5.2 Functions That Work Normally

Description – Functions That Work Normally

Most of the GUS Display functions that can be initiated from a GUS Client in a GUS Remote Cluster work normally, the same as they do when they are initiated from a GUS node. Examples of some of these functions are as follows:

List of Functions

- Executing a left button click on an object scripted with the OnLButtonClick subroutine
- Clicking the SP button on a GUS Faceplate
- Building a new GUS Display (Remote Engineering)
- Calling up another GUS Display
- Calling up a GUS Group Display
- Changing a Parameter Value
- Calling up the GUS Alarm Summary
- Acknowledging alarms from the GUS Alarm Summary
- Calling up a GUS Faceplate
- Changing a Parameter Value using a GUS Faceplate



TIP

This is not a complete list of all the functions that operate identically.

5 GUS Display Functions

5.3 Functions That Work, But Require Additional Actions

5.3 Functions That Work, But Require Additional Actions

Summary

The following TPS operator functions require additional operator actions when executed from a GUS within a GUS Remote Cluster:

- Invoking a Native Window from a GUS Server
- Invoking a Native Window from a GUS Client
- Performing an Area Change

Procedure to invoke a Native Window from a GUS Server

Step	Action
1	Select the Start button and then select Programs > Honeywell TPS > Native Window. The Native Window is displayed.
2	Proceed as follows, if the standard Native Window error message is displayed, due to the Native Window of a GUS Client being connected to this GUS Server: • Disconnect the remote Native Window. • Repeat step 1 above to invoke the Native Window.

Procedure to invoke a Native Window from a GUS Client

Prerequisites

- GUS Server name has been configured.
- Native Window Client has been installed at the GUS Client.
- GUS Server Native Window has been configured to accept remote connections.

Procedure

Step	Action
1	Select the Start button and then select Programs > Honeywell TPS > Native Window. The Native Window is displayed.
2	Proceed as follows if the standard Native Window error message is displayed due to the Native Window already being displayed on a GUS Client connected to this GUS Server: • Determine which GUS Client is currently displaying the Native Window and ask the operator at that GUS to close the Native Window. • Repeat step 1 above to invoke the Native Window.

Procedure to perform an Area Change

Step	Action
1	Notify all the operators currently working on the GUS Clients in the GUS Remote Cluster (by a telephone call for example) to close all their GUS Displays in preparation for an Area Change.
2	Perform the Area Change. A new GUS Picture search path and new key definitions are sent to all the GUS Clients in the GUS Remote Cluster.
3	Notify all the operators who were working on the GUS Clients in the GUS Remote Cluster that the Area Change has been completed and they can now resume operations.
	ATTENTION When a GUS Client connects to the GUS Server, the area definitions are sent automatically to the GUS Client.

5.4 Functions That Are Not Available

Overview

There are also a number of GUS Display functions that do not work when they are executed from a GUS within a GUS Remote Cluster. These functions are as follows:

- Invoking a Native Window Schematic from a GUS Display
- Invoking a GUS Display from a Native Window

5 GUS Display Functions

5.4 Functions That Are Not Available

- Invoking a Cross-screen Display
- PMK Raise/Lower



ATTENTION

All of these functions work on ES-T nodes, even if hosting remote GUS clients.

Invoking a Native Window schematic from a GUS Display

An object in a GUS Display can be scripted to invoke a Native Window schematic through either the InvokeDisplay statement or an Actor. If you execute this script (by clicking on the object, for example), the schematic will not appear. Instead, the error dialog defined in the script error handler will be displayed.

Invoking a GUS Display from a Native Window

If you select the SCHEM key in a running Native Window and enter the name of a valid GUS Display, the display will not appear. Instead, the standard “Schematic Not Found” error message will be displayed.

Invoking a Cross-screen Display

An object in a GUS Display can be scripted to invoke a cross-screen display using a TDC actor. If you execute this script (by clicking on the object, for example), the function will not be executed and a script error handler will be displayed.

PMK Raise/Lower Functions

The Raise/Lower and Fast Raise/Fast Lower functions that are normally available through the PMK scripting object are not available on GUS Remote Displays. If you execute this function, a script error handler will be displayed.

6. Recovering From Hardware Problems

6.1 Hardware Failures

Four Types of Hardware Failures

There are four types of hardware failures that can occur:

- GUS Server failure
- GUS Client failure
- Communications link failure
- GUS Server component failure

6.2 Possible Failure Scenarios

GUS Server or Communications Link failure – Initial Conditions

- GUS Server running
- GUS Client(s) running
- GUS Display(s) running on the GUS Client
- Native Window running on the GUS Client

6 Recovering From Hardware Problems

6.2 Possible Failure Scenarios

GUS Server or Communications Link failure – Failure Indication

	Failure Indication
1	GUS Server (or Communications Link) fails.
2	The communication software detects the failure and causes the GUS Remote Status icon to turn red  and begin blinking, indicating a fatal error.
3	The GUS Display runtime software detects the failure and returns a "Communication Error" to all scripts.
4	The process values on the GUS Display are replaced by the appropriate error indications (for example, a process value is replaced by a "???").
5	The GUS Faceplate, Alarm, and Message OCXs detect the failure and display the appropriate error indications (for example, a process value is replaced by a large, red "X").

GUS Server or Communications Link Failure – Recommended Action

Step	Action
1	Select Programs > Honeywell TPS > GUS Remote Display Status, or Right-click the GUS Remote Displays Status icon. The GUS Remote Displays Status window appears.
2	Review the failure information displayed and take appropriate action.
3	Restart the failed GUS Server by executing the following steps, if required: - Restart the GUS Server by rebooting the GUS station using the standard procedure. - At the GUS Server, the GUS Remote Displays Status window appears, with the GUS Server page displaying a message asking you to load the Native Window. - At the GUS Server (or from any PC using the Remote Native Window), invoke the Native Window. - At the GUS Server, (or from any PC using the Remote Native Window), load the LCNP personality.
4	Restart each GUS Client in the GUS Remote Cluster by executing the following steps: - Log off. - Log back on
5	Reload and rerun the GUS Displays on each GUS Client in the GUS Remote Cluster.

GUS Server Component Failure – Initial Conditions

- GUS Server running
- GUS Client running
- GUS Display(s) running on the GUS Client
- Native Window running on the GUS Client

6 Recovering From Hardware Problems

6.2 Possible Failure Scenarios

GUS Server Component failure – Indication Example

	Failure Indication
1	LCN File Service fails.
2	The communication software detects the failure and causes the GUS Remote Status icon to turn yellow  , indicating a non-fatal error.

GUS Server Component failure – Recommended Action

Step	Action
1	Select Programs > Honeywell TPS > GUS Remote Display Status . Or Right-click the GUS Remote Displays Status icon. RESULT: The GUS Remote Displays Status window appears.
2	Review the failure information displayed and take appropriate action.
3	If the GUS Server has failed, restart it by executing the following steps: - Restart the GUS Server by rebooting the GUS station using the standard procedure. - At the GUS Server, the GUS Remote Displays Status Window appears, with the GUS Server page displaying a message asking you to load the Native Window. - At the GUS Server (or from any PC using the Remote Native Window), invoke the Native Window. - At the GUS Server, (or from any PC using the Remote Native Window), load the LCNP personality.
4	If a GUS Server has failed, execute the following steps on each GUS Client: - Log off. - Log back on - Rerun the GUS Displays

GUS Client Failure – Initial Conditions

- GUS Server running
- GUS Client running
- GUS Display(s) running on the GUS Client.

GUS Client Failure – Failure Indication

	Failure Indication
1	GUS Client fails

GUS Client failure – Recommended Action

Step	Action
1	Restart the failed GUS Client by executing the following steps: • Log off. • Log back on
2	Rerun the GUS Displays.

6 Recovering From Hardware Problems

6.2 Possible Failure Scenarios

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